Project Description

Overview

This is a team project and worth 30% of your final grade. It intends to evaluate your understanding and practical skills of Software Deployment and Evolution with the knowledge of Deployment Fundamentals, Deployment Scripts, Evolution of software and deployment, DevOps. There are four main tasks you are asked to finish in this project.

In this project, you are required to implement a practical application based on the project. The project assessment involves a project brief, the project demonstration, and a final project report (Team-based). This particular project task is to design and construct a DevOps tool chain which is assessed toward 30% of your unit grade.

Always engage in the discussion forums and further announcements for the project-related matters which will be available in Canvas. You are responsible for checking Canvas on a regular basis to stay informed with regard to any updates about the project.

One submission per group. The project requires you to design and construct a DevOps server pipeline. You can select and install the necessary tools of your choice to support a DevOps team using multiple (virtual) servers and infrastructure.

Your Project Application

You have freedom to choose an application that genuinely interests your team (really, pick a problem that **YOU ARE** interested in!). The application is the vehicle for demonstrating your pipeline keep it simple enough that the focus stays on deployment, not development.

Permitted application types:

- **Web app**: frontend (React, Vue, plain HTML/JS) with a backend API (Node, Python, etc.)
- **Desktop app**: Electron, Tauri .NET, or similar, optionally paired with a cloud or hosted backend
- **Hybrid**: any mix of the above, e.g. a desktop client backed by a cloud API

The application does not need to be complex or original. A simple app that is reliably deployed, monitored, and automatically updated through your pipeline is worth far more than an impressive app with a poor pipeline.

Tier Requirements

Each tier builds on the previous. Work through them sequentially and keep a screen recording or screenshots of every sub-task you perform, this evidence is required for your report submission.

Level 1 Pass (50–59%)

Create a simple DevOps pipeline including a code repository server, a CI/build server, a test server, and a production server.

Level 2 Credit (60-69%)

Complete Level 1 and add instrumentation to the production server which can gather stats on server performance.

Level 3 Distinction (70-79%)

Complete Level 2 and deploy a simple application using the DevOps pipeline. Verify that your application works accurately and is accessible.

Level 4 High Distinction (80-100%)

Complete Level 3 and test the automation scripts by changing the source code to trigger a deferred or automatic deployment of the new build.

Suggested Tools (Examples Only)

You are free to use any tools. These are common options; you are not required to use any of them.

Pipeline Stage	Example Options
Source Control	GitHub, GitLab, Bitbucket
CI / Build	GitHub Actions, GitLab CI, Jenkins, CircleCI, AWS CodeBuild, Azure DevOps
Testing	Jest, pytest, JUnit, built-in CI test steps
Hosting / Deployment	Render, Vercel, Fly.io, Railway, AWS EC2, AWS Lambda, Azure App Service, GCP Cloud Run
Database	Neon, Supabase, PlanetScale, AWS RDS, MongoDB Atlas
Monitoring	Render/Vercel dashboards, AWS CloudWatch, Prometheus + Grafana, Datadog (free tier), UptimeRobot

Assessment Requirements

Project Brief

For your project brief submission, the team supposed to identify the tools that you will use to build the DevOps pipeline and three environments. You could use something like Jenkins for CI, build server and test server, and AWS instance for Production server. To maintain your code repository, you could use GitHub or any other collaborative tool you feel comfortable using.

In your project brief submission document, please provide a brief explanation of DevOps server pipeline. Explain each of the tools you have selected to complete your project stating how and why would you use them. Provide a clear explanation justifying your selection of tools.

Project Presentation

Before you present your work, the team must present all levels of attempt. Work on the tasks sequentially keeping a screen of every sub-task you perform to complete the tasks at every level for project report submission. The assessment is due in week 11 and you will be presenting your work in your respective tutorial workshops. For presentation, you can support your work with some presentation slides (4–8 slides). Slides could talk about the selected tools, problems faced and how they were resolved, code demonstration of the deployed project. The presentation will be 15 minutes long followed by Q&A.

Project Report

The team is required to submit a project report providing evidence of completion of the various sub-tasks. The evidence can include screenshots/clear images (labeled with captions), pasted text from console windows and description of what you did and any errors/unresolved problems. Also document how well you were able to complete the project and the lessons learned.